THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The present ages of two brothers are in the ratio 3 : 2. After 4 years the ratio becomes 7 : 5. The present age of the elder brother is: 1

(a) 24 years **(b)** 16 years **(c)** 28 years **(d)** 40 years

2. The population of a town is 50000 and it grows at 10% per annum. After 2 years it will be: 1

(a) 66550 **(b)** 54450 **(c)** 60500 **(d)** 60000

3. The sides of a rectangle are in the ratio $3 : 4$ and its diagonal is 45 cm. The area of the rectangle (in cm²) is: 1

(a) 972 **(b)** 405 **(c)** 1053 **(d)** 126

4. ₹ 4500 is divided among three friends in the ratio 2:3:5. The difference between the largest and the smallest share (in ₹) is: 1

(a) 900 **(b)** 1350 **(c)** 2250 **(d)** 1351

5. A polyhedron has 18 edges and exactly 3 edges meet at every vertex. Its number of faces is: 1

(a) 6 **(b)** 20 **(c)** 12 **(d)** 8

6. The mean of 10 values was found to be 36. Later it was found that 80 was misread as 30. The correct mean is: 1

(a) 31 **(b)** 41 **(c)** 36 **(d)** 86

7. If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 9 **(b)** 7 **(c)** 6 **(d)** 11

8. An isosceles trapezium has parallel sides 10 cm and 22 cm and non-parallel sides 10 cm. Its area is: 1

(a) 160 cm² **(b)** 128 cm² **(c)** 138 cm² **(d)** 256 cm²

9. The largest number to be subtracted from 2577 to make it a perfect square is: 1

(a) 79 **(b)** 80 **(c)** 78 **(d)** 77

10. The number of digits in $2^4 \times 5^7$ is: 1

(a) 8 **(b)** 6 **(c)** 7 **(d)** 9

11. Which of the following numbers is the greatest? 1

(a) 16_{10} **(b)** 14_8 **(c)** 1101_2 **(d)** 32_5

12. In a rhombus ABCD, $\angle ACD = 25°$. Then $\angle ABC$ is: 1

(a) $65°$ **(b)** $130°$ **(c)** $50°$ **(d)** $155°$

13. The number of 3-digit numbers divisible by 8 is: 1

(a) 112 **(b)** 113 **(c)** 111 **(d)** 124

14. The value of $512^2 - 488^2$ is: 1

(a) 24100 **(b)** 500288 **(c)** 576 **(d)** 24000

15. ₹ 1300 is divided in parts proportional to $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. The largest part is: 1

(a) ₹ 260 **(b)** ₹ 300 **(c)** ₹ 600 **(d)** ₹ 400

16. A machine worth ₹ 50000 depreciates at 10% per annum. Its value after 2 years (in ₹) is: 1

(a) 44550 **(b)** 36450 **(c)** 40500 **(d)** 40000

17. **Assertion (A):** The number 4,236 is divisible by 3. 1

Reason (R): A number is divisible by 3 if the sum of its digits is divisible by 3.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

18. **Assertion (A):** $(x + 5)^2 = x^2 + 10x + 25$. 1

Reason (R): $(a + b)^2 = a^2 + 2ab + b^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. **Assertion (A):** The ratios $3 : 4$ and $9 : 12$ are in proportion. 1

Reason (R): Two ratios $a : b$ and $c : d$ are in proportion if they are equal, that is, if $ad = bc$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** 25% of 360 is 90. 1

Reason (R): $x\%$ of a quantity N equals $\frac{x}{100} \times N$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. The sides of two squares are in the ratio 3 : 7. Find the ratio of their perimeters and of their areas. 2

22. A number is increased by 25% and the result is decreased by 10%. Is the final number more or less than the original? By what percent? 2

23. The legs of a right triangle are in the ratio $3 : 4$ and the hypotenuse is 35 cm. Find the legs. 2

24. If $x : y = 3 : 4$, find the ratio $(5x + 2y) : (4x - 1y)$. 2

25. A cube is painted all over and cut into unit cubes. If 48 unit cubes have exactly two painted faces, find the side of the cube and the total number of unit cubes. 2

26. The mean of 6 numbers is 20. When one number is removed, the mean of the remaining 5 numbers is 18. Find the removed number. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A ladder 37 m long, with its foot in a street, reaches a window 35 m high on one side. Without moving its foot it is turned to reach a window 12 m high on the opposite side. Find the width of the street. 3

28. A garrison of 240 men has provisions for 45 days. After 15 days 60 men leave. For how many more days will the remaining provisions last for the men who stay? 3

29. Every face of a polyhedron is a triangle and it has 32 faces. Find the number of edges and vertices and verify Euler's formula. 3

30. A class of 40 students has mean mark 62. After rechecking, a score of 38 was corrected to 80. Find the new mean. 3

31. The sum of a two-digit number and its reverse is 110 and the digits differ by 4. Find the number (tens digit larger). 3

OR

The perimeter of a rectangular plot is 160 m and its length is 3 times its breadth. Find its area and the cost of fencing at ₹ 15 per metre. 3

32. A solid cube of side 6 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

OR

Light travels 3×10^8 m in one second. How far does it travel in 2 minutes? Give the answer in scientific notation. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** The foundation stone of a school in Nagpur reads MCMLXXXV, and the plaque on the new building reads XXIV. 5

(i) Convert MCMLXXXV to a Hindu-Arabic number.

(ii) Convert XXIV to a Hindu-Arabic number.

(iii) Write 2024 in Roman numerals.

(iv) The school year is 2024; find the age of the school and write it in Roman numerals.

34. **Case study.** A carpenter makes a parallelogram-shaped window frame ABCD with $\angle A = 70^\circ$, $AB = 80$ cm and $BC = 50$ cm. The diagonals AC and BD meet at O, with $AC = 96$ cm and $BD = 60$ cm. 5

(i) Find $\angle C$.

(ii) Find $\angle B$.

(iii) Find the perimeter of the frame.

(iv) Find OA and OB, and state the property used.

35. The mean of 15 numbers is 65, the mean of 25 numbers is 15 and the mean of 10 numbers is 25. Find the mean of all 50 numbers. 5

36. Prove that $(2n+3)^2 - (2n-3)^2 = 24n$ and use it to evaluate it for $n = 4$. 5

OR

The area of a rhombus is 216 cm² and one diagonal is 18 cm. Find the other diagonal, the side, and the altitude. 5

37. Find the smallest cube number that is divisible by each of 4, 6 and 9. 5

OR

Body A has mass 6×10^{24} kg and body B has mass 3×10^{22} kg. Which is heavier? How many times as heavy is it? Find the total mass in scientific notation. 5

38. Subtract $1222_5 - 341_5$ in base 5. Verify your answer in decimal. 5

OR

In a parallelogram ABCD, AB $= (3x + 4)$ cm, CD $= (2x + 9)$ cm and AD = 12 cm. Find x, the lengths of AB and CD, and the perimeter. 5

END OF QUESTION PAPER **Total: 80 Marks**

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